

Week-2 Package & Peer Signal Review

1. WEEK 2 PACKAGE — SUMMARY

Day	Focus	Key Outcome
8	Video/Signal Quality	Motion negligible (<5px); found and filtered a non-transmitter artifact in the 1-LED video; ambient light stable
9	ROI/Signal Plan (Mentor Review 3)	Fixed, area-filtered ROI approved per video; 2.826 frames/bit flagged as a sync risk before any data showed it
10	Frame & Intensity EDA	SNR excellent (57x-318x) — but transition timing is frame-quantized, not bit-period-aligned. First hard evidence of the sync risk
11	Signal Preprocessing	Signal V1 built: fixed ROI, light median-3 denoise, per-video minmax normalization. Found a ramp-up transient at active-window start
12	Signal Pipeline Review (Mentor Review 4)	Signal V1 presented and frozen; denoise/normalize choices defended with Day 10-11 evidence, not asserted
13	Synchronization Analysis	Two independent methods (autocorrelation, interval residual minimization) agree: per-frame clock recovery does not work at 2.826 frames/bit. Rolling-shutter row-level extraction flagged as the priority for Week 3

2. THE THROUGHLINE

Week 2 moved from 'the signal looks clean' to 'the signal is clean but under-sampled'

That's a real, specific, falsifiable finding, not a vague 'more work needed', and it points directly at what Week 3 must solve: get finer time resolution than one sample per frame, most likely via rolling-shutter row-level extraction, before a decoder can work.

3. DIFFICULT VIDEO SEGMENTS — PREPARED FOR DISCUSSION

Concrete segments worth raising with a peer, since 'difficult' is more useful specific than general:

- 1-LED video, ~frame 5850-13000 region: the secondary bright artifact (Day 8) appears and disappears — worth asking a peer how they're handling non-transmitter bright regions in their own dataset, if at all.
- All 4 videos, active-window start (first ~150-300ms): the ramp-up transient found Day 11, did a peer's dataset show a comparable warm-up period, or is this specific to this capture setup?

- The 4-LED video specifically: interval residual-minimization (Day 13) hit the edge of the search range (6.0ms) rather than a clean interior minimum — worth a second opinion on whether that's a real signal or a search range artifact.

4. NEXT STEP

Week begins actual decoding: threshold baseline first , then improved synchronization work that directly follows up on the Day 13 finding and whatever the peer discussion above adds to it.